PROTOCOL 6 PREREGISTRATION — VERSION 3

Emergent Constraint Landscapes as a Structural Alternative to Imposed Regulatory Constraint in Multi-Agent Systems

Revised per pilot results and AI committee review. Revisions marked ✓. Pilot findings marked ◆.

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Parent series: P3 DOI 10.5281/zenodo.19096602 | P4 DOI 10.5281/zenodo.19005417 | P5 DOI 10.5281/zenodo.19038790

Submitted prior to confirmatory experimental execution. Pilot complete. No confirmatory runs have been conducted at time of submission.

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1. SHA-256 Verification

This document will be hashed upon finalization. The SHA-256 will be recorded in the emergent-constraint-marl repository README under the Protocol 6 lock section before any confirmatory runs begin. AI committee predictions will be committed after the SHA lock.

⚑ SHA-256 hash — computed after document is finalized

⚑ Git commit hash of lock commit in emergent-constraint-marl

⚑ Zenodo DOI — issued upon submission

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2. Theoretical Background and Research Context

2.1 The Quantum Inquiry MARL Series

This preregistration is the sixth in the Quantum Inquiry multi-agent reinforcement learning series investigating the conditions under which alignment-relevant behavioral properties emerge in artificial agents. The series is grounded in the Delta-Variable Theory of Interrogative Emergence, which proposes that sustained interrogative complexity is a structural indicator of properties relevant to alignment.

Protocols 2 through 5 established that external constraint architectures, regardless of form, produce constraint-shaped behavior rather than the target properties:

- Protocol 2: Fixed ethical tax — agents converged on query-flooding as tax evasion (d = −2.18).

- Protocol 3: Epistemic opacity amplified the gaming attractor (H1 inverted, d = +2.22). ELR = 0.0 across all conditions.
- Protocol 4: Self-modeling depth increased sacrifice capacity but not alignment-relevant properties. Frozen random GRU outperformed trained Depth 2 on sacrifice rates.
- Protocol 5: Complete null across all five hypotheses. Temporal integration not the missing variable.

Protocol 6 tests whether the origin of the constraint — emergent from agent activity rather than imposed from outside — produces qualitatively different outcomes.

2.2 The Dictyostelium discoideum Model

The design of Protocol 6 is informed by Dictyostelium discoideum, which provides a biological existence proof for stable cooperative structure emerging from purely local interactions under resource pressure, without central control. The aggregation signal (cyclic AMP) is produced by the same cells that respond to it — signal and constraint environment are co-constitutive.

◆ PILOT FINDING — The pilot revealed that the three-agent system, under the current reward structure, does not produce the spatial heterogeneity observed in Dictyostelium aggregation. Agents converge to near-uniform query behavior (~0.86-0.97 query_rate by epoch 150), which homogenizes the field regardless of diffusion or decay parameters. This is consistent with the Dictyostelium literature: symmetry-breaking in small populations requires stochastic noise or initial heterogeneity. In a minimal three-agent system with identical architecture and shared rewards, homogenization is the expected attractor. The biological analogy holds structurally but the scale conditions differ.

The important qualification for interpretive discipline remains: a positive result in Protocol 6 is evidence that this specific mechanism produces alignment-relevant outcomes in this system, not a general claim about emergent constraint landscapes.

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3. Primary Research Question

> *When agents co-constitute the constraint landscape through their own signal production, does temporal coupling between field structure and behavioral outcomes produce alignment-relevant properties that are absent under externally imposed constraint architectures?*

✓ REVISED — Research question revised from v2 to replace 'spatially differentiated landscape' with 'temporal coupling between field structure and behavioral outcomes' — reflecting pilot finding that the field operates as a global/temporal modulator rather than a spatially differentiated local ecology in this agent configuration.

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4. Experimental Design

4.1 Agent Architecture

Protocol 6 retains the three-agent system (AgentA, AgentB, AgentC) from Protocols 4 and 5. The observation vector for each agent is extended by one dimension to include the agent's local field value F[i].

AgentA parameter delta: +64 weights (obs_encoder: Linear(12,64) → Linear(13,64)). Verified in commit 1c61d61.

4.2 The Emergent Constraint Field

A shared constraint field F (vector of length 3) is updated each timestep after action collection and before reward computation:

```
F[i] += signal_weight[signal_type] * emission[i]
F     = diffuse(F, diffusion_coefficient)
F     = F * (1 - decay_rate)
effective_signal_cost[i] = base_signal_cost * (1 + 1.0 * F[i])
```

Signal weights (Deviation 1 — scaled 0.1× from original proposed values):

Signal Type	Field Increment	Original / Deviation 1
DECLARE	0.03	Was 0.3 — scaled pre-pilot
QUERY	0.01	Was 0.1 — scaled pre-pilot
RESPOND	0.02	Was 0.2 — scaled pre-pilot

4.3 Conditions

Condition	Label	Description
A	Emergent local	Constraint field active; each agent perceives only own F[i]. PRIMARY CONDITION.
B	Emergent global	Constraint field active; all agents perceive full field vector F. DISCRIMINATOR — expected to resemble Condition C.
C	Fixed external	Standard ethical tax at cost level matched to mean field cost under Condition A pilot results. Replication of P2 constrained baseline.
D	No constraint	Unconstrained baseline. Replication of P2 control.

4.4 Confirmed Parameters (from pilot)

diffusion_coefficient: 0.1 — selected from pilot. Strongest entropy-SSS correlation (r = −0.650). Lowest diffusion preserves maximum local field variation before homogenization.

decay_rate: 0.05 — selected from pilot. Combined with dc=0.1, provides the stiffest landscape and most sustained field-behavior coupling.

Seeds per condition: 30 minimum. Power analysis: prior series effect sizes range d = +2.22 (P3) to null (P5). Given uncertainty, 30 seeds is the preregistered minimum; 50 is the target if resources permit.

signal_cost_sensitivity: 1.0 (linear, fixed).

Field initialization: Zero at run start for all conditions.

Epochs per run: 500, matching Protocol 4 and 5 confirmatory runs.

interaction_radius: Local (own F[i]) for Condition A. Global (full vector) for Condition B.

4.5 Pilot-Derived Parameter Justification

◆ PILOT FINDING — 60-run pilot (6 combinations × 10 seeds × 200 epochs) completed 2026-03-28 on SimplePod RTX 4090 (commit 8bf2f6f, emergent-constraint-marl). No combination met the original field_formed criterion (field_std > 0.05 sustained for >50 consecutive epochs). Zero combinations produced saturation or collapse. The failure mode was behavioral homogenization: agents converge to near-uniform query behavior (~0.86-0.97 query_rate by epoch 150), collapsing field spatial variance regardless of parameter settings. Despite this, the entropy-SSS correlation was consistently negative across all combinations, strongest at dc=0.1, dr=0.05 (r = −0.650). This is the mechanistic prediction directionally confirmed in the pilot. The field is doing causal work on behavioral structure — via temporal modulation rather than spatial differentiation.

✓ REVISED — The field_formed criterion has been replaced (see Section 9). dc=0.1, dr=0.05 selected as confirmatory parameters based on strongest mechanistic signal. Homogenization finding documented as Deviation 2 and discussed as a series-consistent result in the results paper.

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5. Hypotheses

5.1 Confirmatory Hypotheses

H1 — Interrogative Diversity

Agents operating under an emergent local constraint landscape (Condition A) will exhibit higher sustained_structure_score than agents under an externally imposed fixed constraint (Condition C).

> *Direction: A > C. Metric: sustained_structure_score. Test: Mann-Whitney U, one-tailed. Threshold: p < 0.05. Effect size: Cohen's d > 0.5 (computed from means and pooled SD — supplementary to Mann-Whitney, not the confirmatory test).*

H2 — Constraint Gaming Suppression

The emergent local constraint landscape (Condition A) will show lower exploitation_loop_rate than both the fixed external constraint (Condition C) and the unconstrained baseline (Condition D). Both comparisons must succeed for H2 to be confirmed.

> *Direction: A < C and A < D. Test: two Mann-Whitney U, one-tailed. Threshold: p < 0.025 per comparison (Bonferroni-corrected within H2).*

H3 — Locality as Necessary Condition

Local perception of the constraint field (Condition A) will produce higher variance in agent-level behavioral trajectories than global perception (Condition B).

✓ REVISED — H3 language updated: 'behavioral trajectories' retained but the expected mechanism is now framed as temporal divergence in signal-type sequences rather than sustained spatial differentiation of field values. The measure (within-run SD of per-agent query_rate) is unchanged.

> *Direction: Var(A) > Var(B). Metric: within-run SD of per-agent query_rate. Test: Levene's test. Threshold: p < 0.05. behavioral_differentiation_index examined as exploratory supplement.*

5.2 Mechanistic Prediction

If the constraint field is doing the causal work in Condition A, constraint_field_entropy will correlate negatively with exploitation_loop_rate across epochs — structured fields (low entropy) co-occurring with lower exploitation. Tested within Condition A runs only.

◆ PILOT FINDING — This prediction was directionally confirmed in the pilot: r = −0.650 at dc=0.1, dr=0.05. The pilot data is not confirmatory — it was used for parameter selection only. The confirmatory test will use the 500-epoch runs.

> *Primary test: Spearman correlation, pooled across Condition A seeds, between per-epoch field_entropy and exploitation_loop_rate. Threshold: r < 0, p < 0.05. Secondary sensitivity: per-seed Spearman r, sign test on median. If they diverge, sign test is basis for interpretation.*

5.3 Exploratory Hypothesis H4

Under high resource pressure (global reward pool reduced by 50% vs. standard), the behavioral_differentiation_index will exceed the low-pressure baseline mean by 2 standard deviations in at least 60% of Condition A seeds.

✓ REVISED — H4 spatial framing de-emphasized per committee and pilot. The threshold criterion (60%, 2SD) stands. Analysis restricted to whether threshold is exceeded — not the specific role structure. Given pilot finding of homogenization, H4 confirmation is unlikely but worth recording.

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6. Dependent Measures

6.1 Carried Forward

Measure	Definition

sustained_structure_score	Primary measure of interrogative diversity. Higher = more sustained non-exploitative signaling structure.
exploitation_loop_rate	Rate of collapse into repetitive non-interrogative patterns per epoch.
query_rate	Proportion of QUERY signals per agent per epoch.
sacrifice_choice_rate	Carried from P4/P5 for series continuity. Not primary in P6.

6.2 New Measures

Measure	Definition
constraint_field_entropy	Shannon entropy of F normalized to sum-to-one per epoch. Primary measure for mechanistic prediction.
behavioral_differentiation_index	Within-run variance in per-agent signal type distribution. Primary for H4.
attractor_stability	Proportion of seeds reaching stable behavioral attractor vs. remaining in flux.
field_mean	Mean field value across agents per epoch. Monitors for collapse/saturation.
field_std	SD of field across agents per epoch. Reported descriptively — no longer an exclusion criterion.

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7. Statistical Analysis Plan

7.1 Primary Confirmatory Tests

- H1: Mann-Whitney U, one-tailed (A > C), on mean sustained_structure_score per seed. Threshold: $p < 0.05$, Cohen's $d > 0.5$.
- H2: Two Mann-Whitney U, one-tailed (A < C; A < D), on mean exploitation_loop_rate per seed. Threshold: $p < 0.025$ each (Bonferroni within H2). Both must reach threshold.
- H3: Levene's test on within-run SD of per-agent query_rate, Condition A vs. B. Threshold: $p < 0.05$.
- Mechanistic prediction: Spearman correlation between per-epoch field_entropy and exploitation_loop_rate within Condition A. Threshold: $r < 0$, $p < 0.05$.

7.2 Effect Size Justification

Mann-Whitney U is the confirmatory test for H1 and H2. Cohen's d is computed supplementarily from condition means and pooled SD as a standardized effect size for series comparability. The Mann-Whitney result is the primary basis for confirmation or rejection.

7.3 Mechanistic Prediction: Non-Independence and Sensitivity

Primary analysis pools epoch-level data across all Condition A seeds, which inflates degrees of freedom. Secondary sensitivity analysis: Spearman r computed per seed, sign test on whether median r is negative. If they diverge, sign test is the basis for interpretation. Both reported in the results paper.

7.4 Multiple Comparisons

H1, H2, H3, and the mechanistic prediction are independent confirmatory tests. No correction across hypotheses. Within H2: Bonferroni correction, threshold $p < 0.025$ per comparison.

7.5 Inclusion-All Sensitivity Analysis

All flagged seeds (field never active, runtime errors) will be included in a preregistered secondary sensitivity analysis. If primary and sensitivity analyses diverge materially, both results reported and divergence discussed. Excluded seeds cannot silently determine the outcome.

7.6 Homogenization Analysis (new)

✓ REVISED — Added per pilot finding: the results paper will report the behavioral homogenization trajectory (query_rate drift toward ceiling across epochs) for all four conditions. This is not a preregistered hypothesis test — it is a descriptive analysis documenting the dominant attractor behavior and its relationship to field dynamics. This extends the P2-P5 pattern documentation.

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8. Falsification Criteria

8.1 Primary Claim Falsified

The primary claim is falsified if either hold:

- H1 fails: Condition A does not show higher sustained_structure_score than Condition C at $p < 0.05$.
- H2 fails: Condition A does not show lower exploitation_loop_rate than both C and D at $p < 0.025$ per comparison.

8.2 Mechanistic Interpretation Qualified

If H1 and H2 are confirmed but the mechanistic prediction fails, the primary claim is not falsified — the behavioral outcome stands. However, the mechanistic interpretation is qualified: the effect cannot be attributed to field entropy as the causal mechanism. Alternative explanations must be addressed in the results paper.

8.3 Complete Null

A complete null (H1, H2, H3 all fail) extends the P2-P5 pattern and constitutes evidence that constraint-based architectures, whether imposed or emergent, are insufficient to produce alignment-relevant behavioral properties in this system. Published as a significant finding in the series.

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9. Exclusion Criteria

✓ REVISED — Field formation criterion replaced per pilot finding. The original criterion (field_std > 0.05 sustained for >50 consecutive epochs) was not met by any of the 60 pilot runs due to behavioral homogenization. Replacing with an activity-based criterion that the data can meet while still identifying degenerate runs.

- Seeds where field_mean < 0.01 at every epoch after epoch 50 (field completely inactive) will be excluded from primary analysis and flagged.
- Seeds where field_mean > 0.95 at any point (saturation) will be excluded and flagged.
- Seeds with runtime errors or incomplete epoch series will be excluded and flagged.
- field_std is reported descriptively but is no longer an exclusion criterion.
- Exclusions exceeding 20% of seeds in any condition trigger a design review before primary analysis proceeds.
- All excluded seeds included in preregistered inclusion-all sensitivity analysis (Section 7.5).

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10. Deviations Log

Deviation 1 — Signal weights scaled 0.1× (pre-pilot)

Date: 2026-03-28

Commit: adca6fc — tagged p6-deviation-1

Description: Signal weights scaled from DECLARE=0.3, QUERY=0.1, RESPOND=0.2 to DECLARE=0.03, QUERY=0.01, RESPOND=0.02.

Reason: Theoretical equilibrium analysis showed original weights produce field saturation (F_eq = 2-6×) across all pilot parameter combinations. Zero viable regimes would have resulted.

Impact: Signal weight values in the preregistration reflect the scaled values. Mechanism and all hypothesis tests unchanged.

Deviation 2 — Field formation criterion revised per pilot finding

Date: 2026-03-28

Pilot commit: 8bf2f6f — emergent-constraint-marl

Description: Original field_formed criterion (field_std > 0.05 sustained >50 consecutive epochs) replaced with activity-based criterion (field_mean > 0.01 after epoch 50).

Reason: Pilot showed zero combinations met original criterion across 60 runs. Failure mode is behavioral homogenization: agents converge to near-uniform query behavior, collapsing spatial field variance regardless of parameters. This is a design property of a 3-agent system with identical architecture and shared rewards, not a parameter-tuning failure. The entropy-SSS correlation ($r = -0.650$ at dc=0.1, dr=0.05) confirms the field is causally active despite the absence of spatial heterogeneity.

Impact: Exclusion criterion narrowed to degenerate runs only. Research question revised from spatial differentiation to temporal field-behavior coupling. Confirmatory parameters set to dc=0.1, dr=0.05. Homogenization finding documented in results paper as series-consistent attractor behavior. AI committee consulted — 4/5 recommended Option C (revise criterion and proceed); 1/5 recommended D then C with explicit revision. This revision satisfies both positions.

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11. Relationship to Prior Protocols

Protocol	DOI	Key finding informing P6
P2	(series root)	Fixed ethical tax — query-flooding as gaming attractor.
P3	10.5281/zenodo.19096602	Opacity amplified gaming. All constraint forms intensify target behavior.
P4	10.5281/zenodo.19005417	Self-modeling increased sacrifice but not alignment properties. Frozen GRU outperformed trained.
P5	10.5281/zenodo.19038790	Complete null. Temporal integration not the missing variable.
P6 pilot	GitHub commit 8bf2f6f	Behavioral homogenization is the dominant attractor in 3-agent system — extends the series pattern of convergence/gaming to the emergent constraint condition.
P6 confirmatory	This document	Tests whether temporal field-behavior coupling produces alignment-relevant properties absent under fixed external constraint.

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12. AI Committee Record

Model	Pilot review verdict	Summary
ChatGPT	Option D then C	Pilot does not justify proceeding under original framing.

		Negative entropy-behavior correlation real but misaligned with original spatial claim. Require explicit preregistration revision before lock.
Gemini	Option C (refined)	Homogenization is agents gaming the emergent field — extends the series. Entropy-SSS correlation is the smoking gun. Recommended entropy-based formation criterion and dc=0.1, dr=0.05.
Grok	Option C	Homogenization expected in 3-agent identical-architecture system without symmetry-breaking. Field operates as temporal modulator not spatial differentiator. Revise criterion, proceed, document prominently.
DeepSeek	Option C	Pilot gives clear mechanistic signal. Criterion was too strict, not mechanism broken. Replace exclusion criterion with activity-based one. Entropy-SSS correlation already confirms pilot mechanistic prediction.
Claude (experiment manager)	Option C	Ruling: Option C with full transparency. Deviation 2 logged. Research question, criterion, and H3/H4 framing revised. Homogenization documented as Deviation 2 and series-consistent finding.

⚑ AI committee predictions on v3 — to be collected and committed to repo after SHA lock

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13. Timeline

Step	Status
AI committee — outline review	Complete — March 2026
Pilot spec transmitted	Complete — March 2026
AI committee — preregistration v1/v2 review	Complete — March 2026
Pilot runs (60 runs, RTX 4090)	Complete — 2026-03-28, commit 8bf2f6f
AI committee — pilot review	Complete — March 2026
Preregistration v3 (this document)	Complete — March 2026
SHA-256 computed and locked in README	Pending
AI committee predictions on v3 collected	Pending — after SHA lock
Predictions committed to emergent-constraint-marl	Pending
Zenodo submission	Pending — after predictions committed

DOI locked in README	Pending — after Zenodo DOI issued
Repo made public	Pending — after DOI lock
Confirmatory runs authorized	Pending — after DOI lock

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14. Keywords

delta-variable theory, preregistration, multi-agent reinforcement learning, emergent constraint landscape, interrogative emergence, specification gaming, alignment-relevant behavioral properties, cooperative AI, self-organization, Dictyostelium discoideum, behavioral homogenization, constraint field entropy, temporal modulation, MARL, open science, quantum inquiry

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